AWS Load Balancer – Practical Implementation

What is a Load Balancer?

A **Load Balancer** is an AWS service that automatically distributes incoming user requests across multiple EC2 instances. Its main role is to ensure that no single server is overloaded and that the application remains available even if one server fails. In simple terms, a load balancer works like a **traffic controller** that decides which server should handle each request.

Why Do We Need a Load Balancer?

Without Load Balancer

- All traffic goes to one EC2 instance
- Server may slow down or crash during high traffic
- Application downtime may occur
- Poor user experience

With Load Balancer

- Traffic is shared between multiple EC2 instances
- If one server fails, traffic is redirected to another server
- High availability and reliability

Real-World Example

If 1,000 users try to access a website at the same time:

- **Without Load Balancer:** One server handles all requests and may crash
- **With Load Balancer:** Requests are divided among multiple servers smoothly

Types of Load Balancers in AWS

1. Application Load Balancer (ALB)

- Works at the application layer (HTTP/HTTPS)
- Best for web applications
- Supports routing based on URLs (example: `/login`, `/admin`)
- Most commonly used load balancer

2. Network Load Balancer (NLB)

- Works at the transport layer (TCP/UDP)
- Extremely fast with low latency
- Used for high-performance applications

3. Classic Load Balancer (CLB)

- Older version of load balancer
- Limited features
- Not recommended for new applications

Main Components of a Load Balancer

1. Listener

- Listens for incoming traffic
- Common ports: 80 (HTTP), 443 (HTTPS)

2. Target Group

- Group of EC2 instances
- Load balancer forwards traffic to these instances

3. Health Check

- Regularly checks whether EC2 instances are working properly
- Unhealthy instances do not receive traffic

4. Load Balancer DNS Name

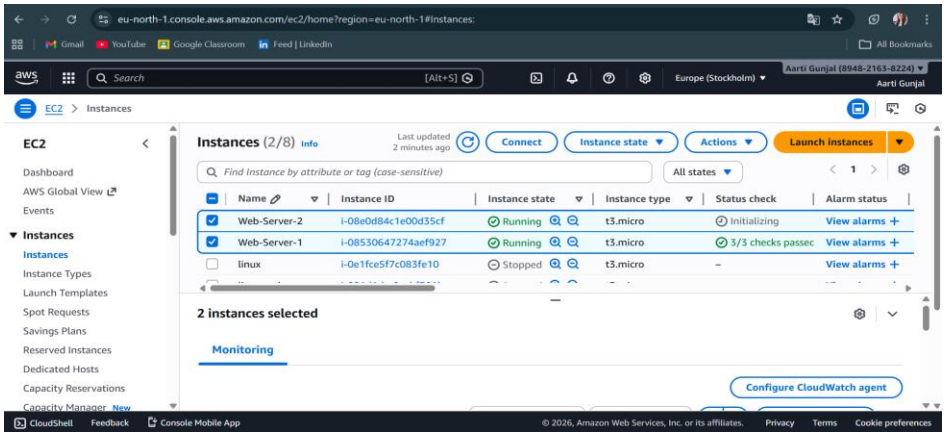
- Public URL provided by AWS
- Used to access the application

Step 1: Launch Two EC2 Instances

In this step, two EC2 instances are created to host the same web application.

Steps

1. Log in to AWS Management Console
2. Open **EC2 Dashboard**
3. Click **Launch Instance**
4. Create two instances with names:
    - Web-Server-1
    - Web-Server-2



Step 2: Deploy Website on Both EC2 Instances

Step 2.1: Connect to EC2 Instance

- Connect using SSH or MobaXterm
- Login user: `ec2-user`
- Use your `.pem` key file

Step 2.2: Update System

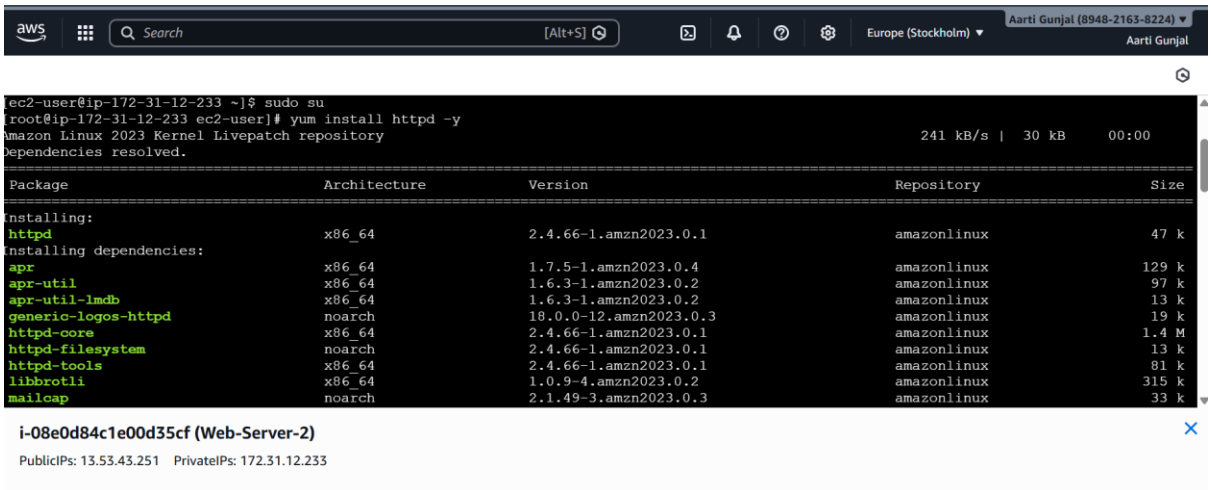
```
sudo dnf update -y
```

Step 2.3: Install Apache Web Server

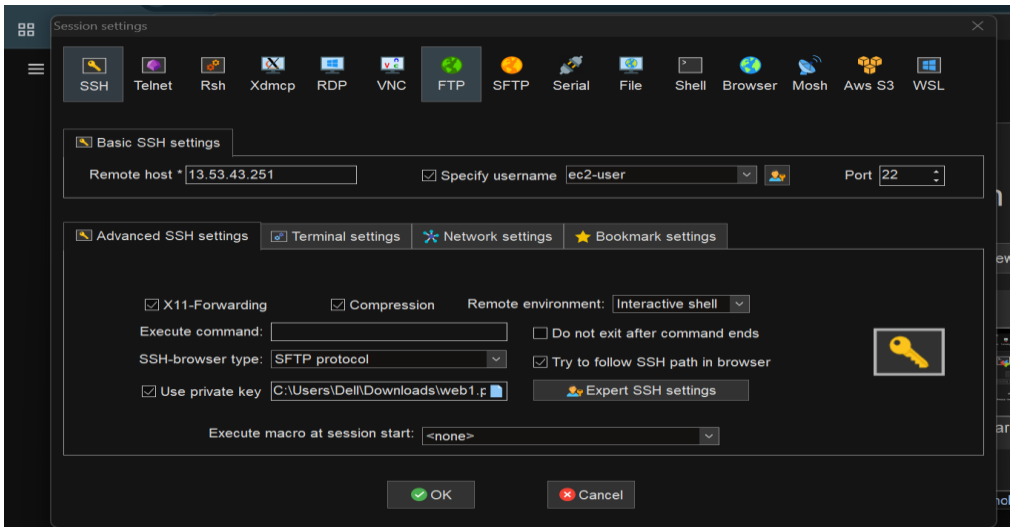
```
sudo dnf install httpd -y
sudo systemctl start httpd
sudo systemctl enable httpd
```

Step 2.3: Restart Apache

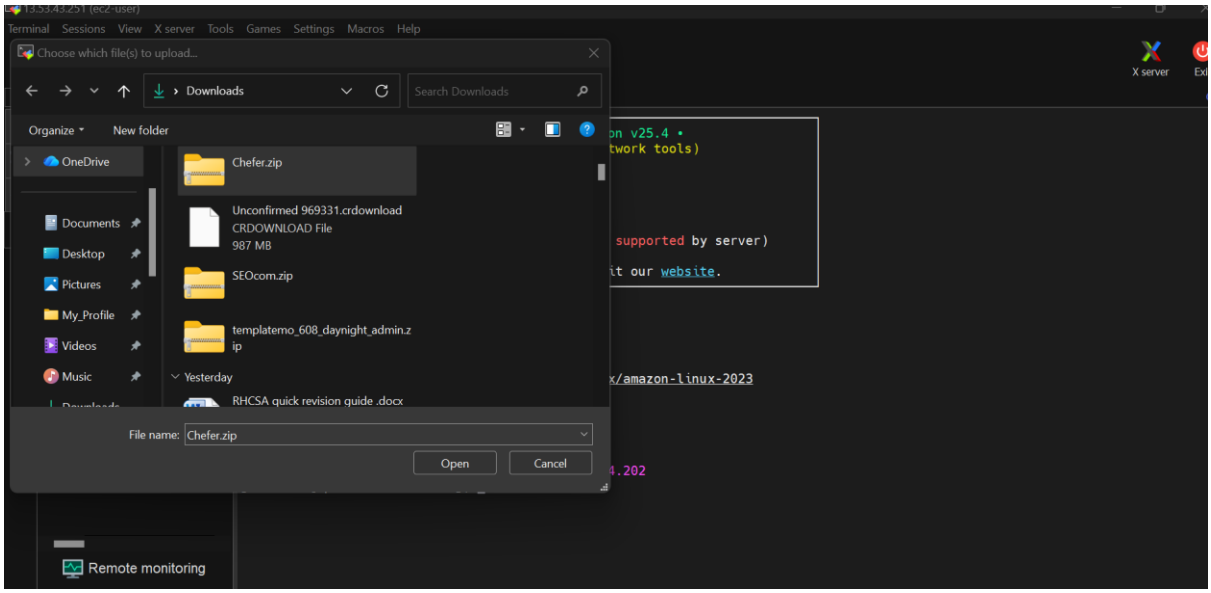
```
sudo systemctl restart httpd
```

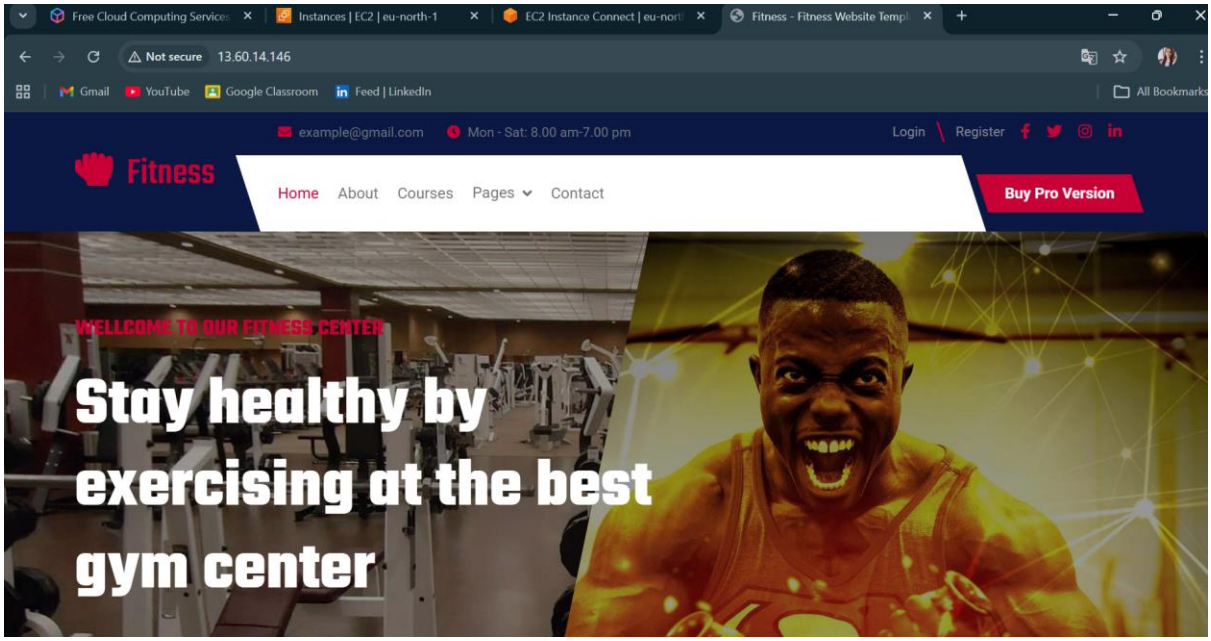


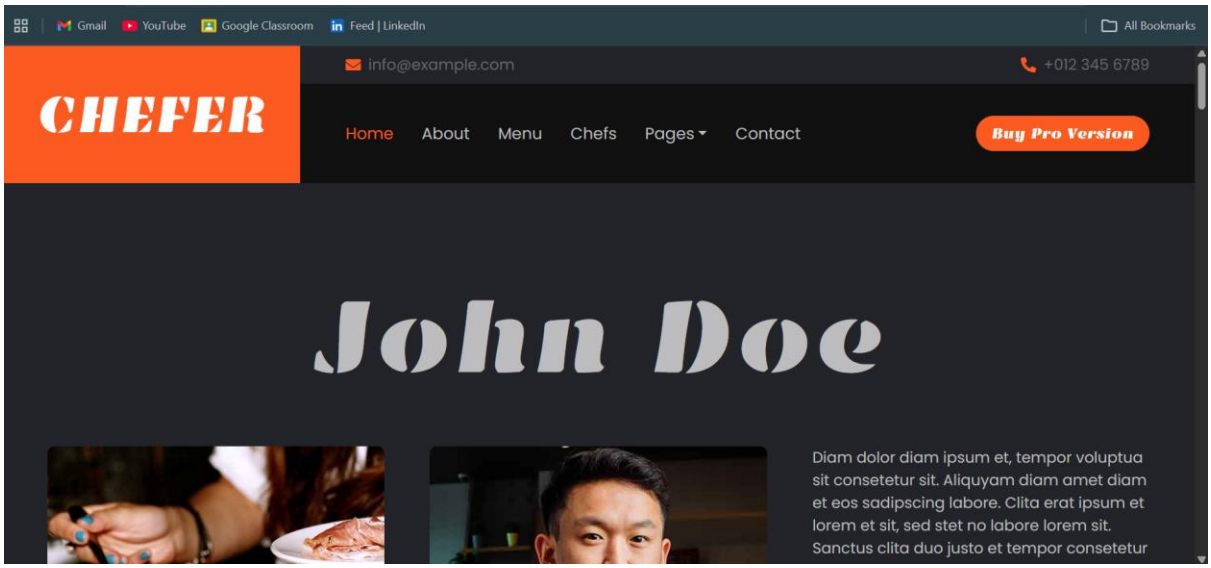
17. Open MobaXterm and create the SSH sessions for both the servers



19. Download the Html Templates and upload it on MobaXterm



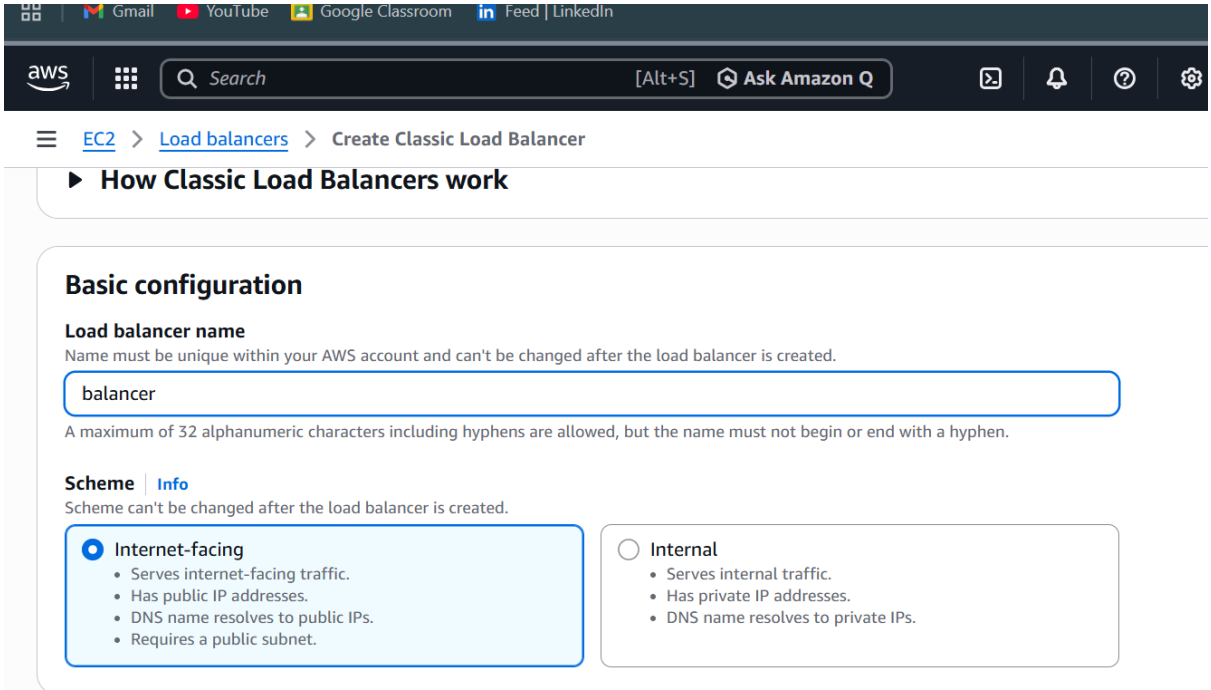




Step 3: Create Load Balancer

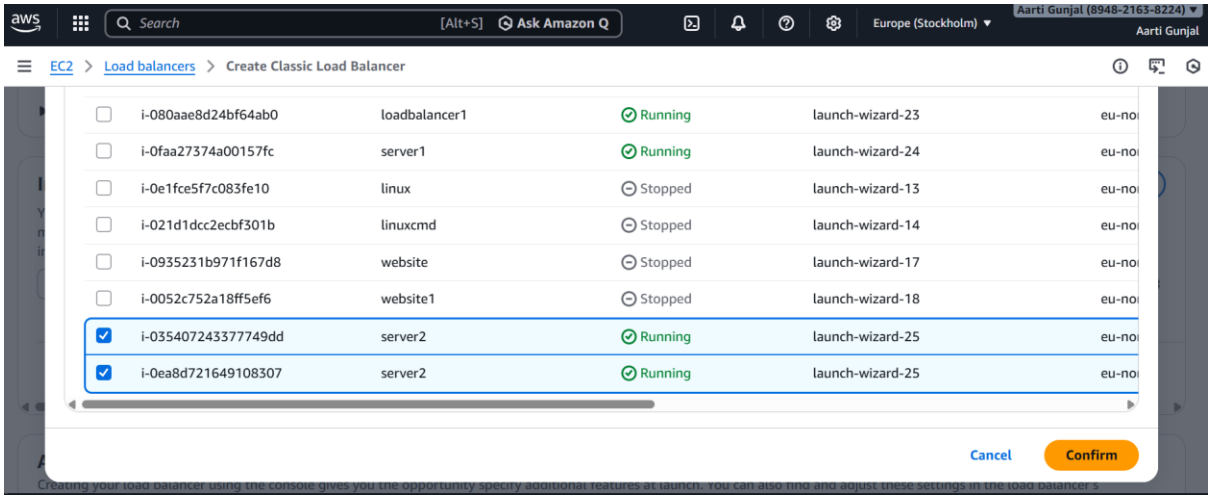
Step 3.1: Open Load Balancer Section

1. Go to EC2 Dashboard
2. Click **Load Balancers**
3. Click **Create Load Balancer**



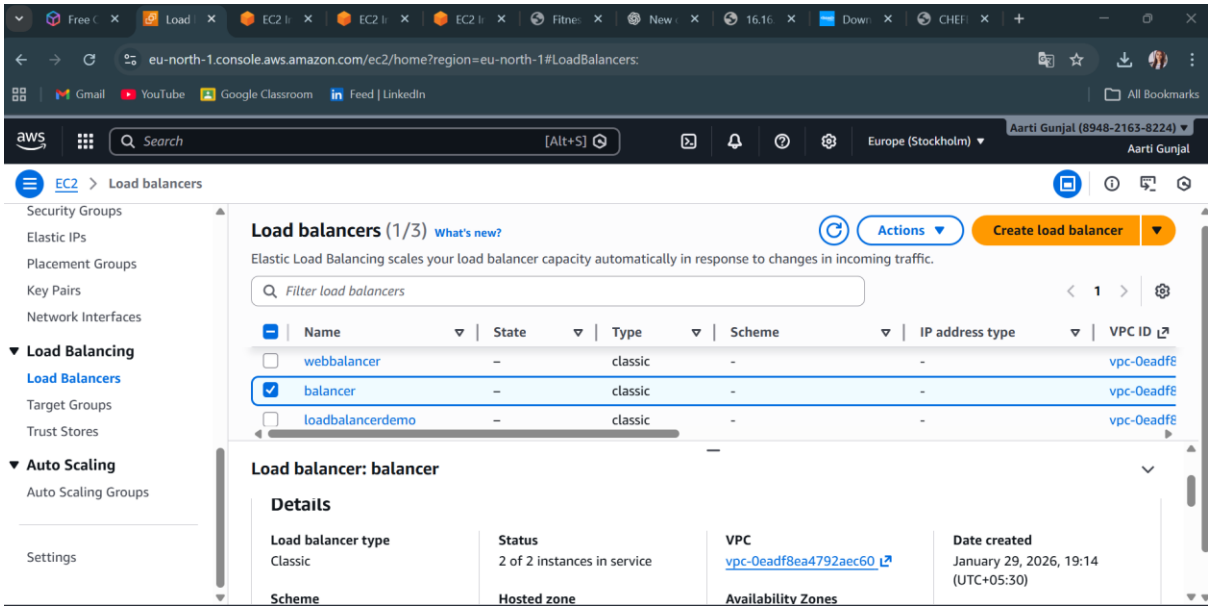
Step 3.2: Select Load Balancer Type

- Choose **Classic Load Balancer**
- Click **Create**



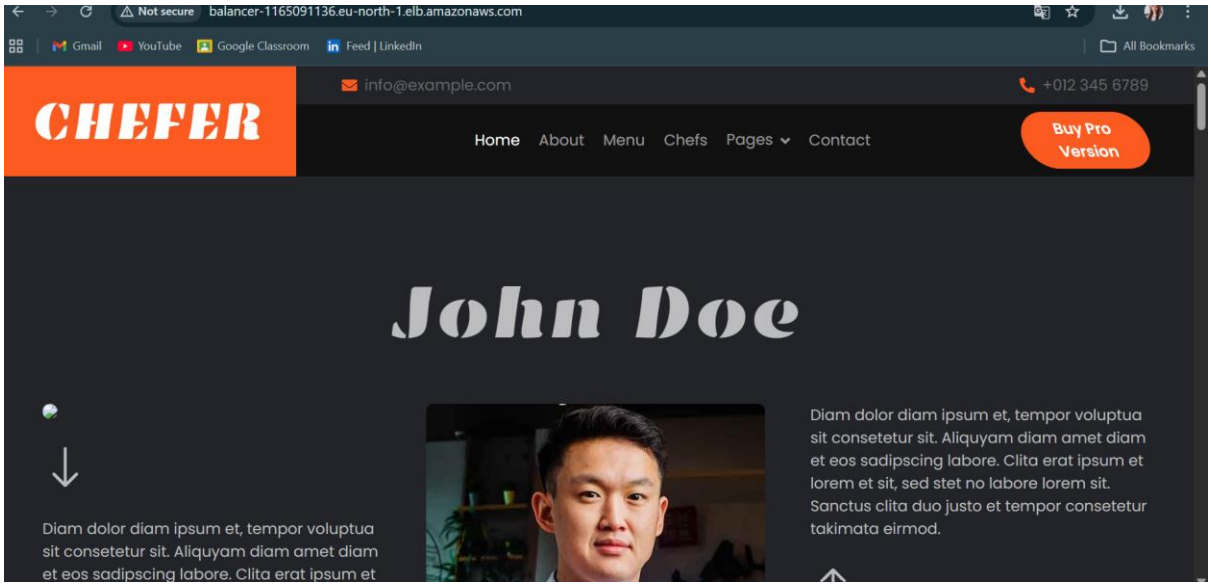
Step 3.3: Configure Load Balancer

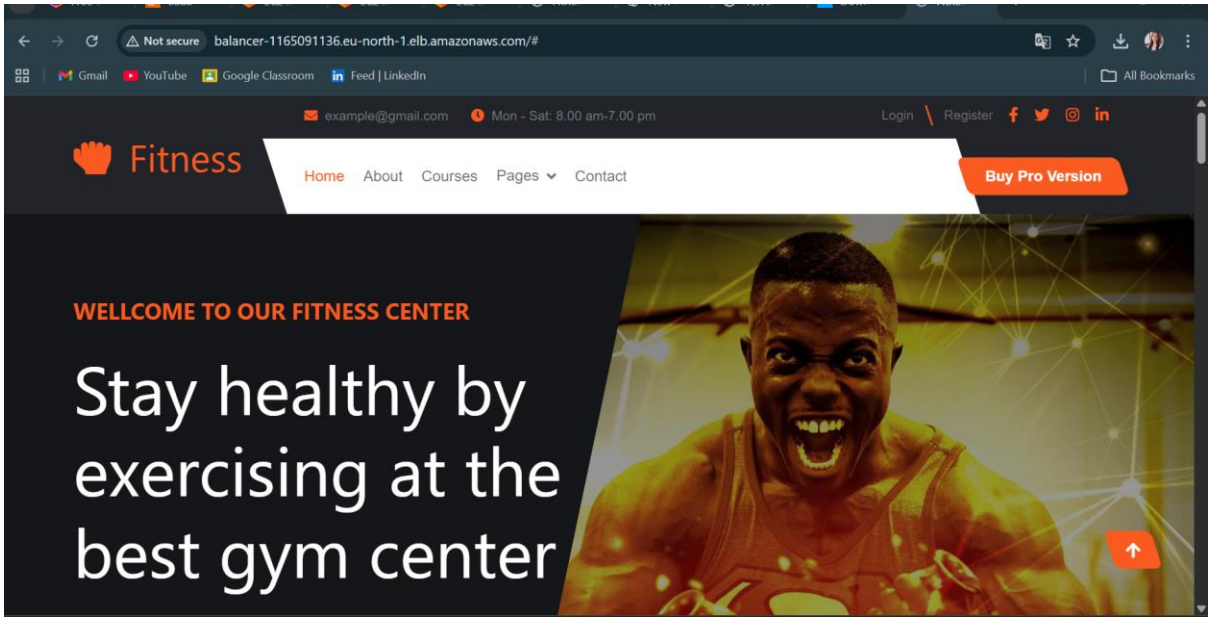
- Enter Load Balancer Name
- Select both EC2 instances:
  - Web-Server-1
  - Web-Server-2
- Review settings and create



Step 4: Test Load Balancer

1. Copy the Load Balancer DNS name
2. Paste it into a browser
3. Refresh the page multiple times
4. Traffic will switch between EC2 instances

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